Timothy **Jan**

✉ timothy_jan@berkeley.edu
909-413-2769
✉ 15701 Willow Glenn Dr, Chino Hills, CA, 91709

Profile

Ambitious and adaptable Physics, Math, and Public Policy student at UC Berkeley. Interested in solid-state, particle, and nuclear physics. Additional interests include semiconductors, IC circuit design, data analysis, and venture capital.

Personal information

Languages: **English** (Native), **Chinese** (Native), **Spanish** (Elementary)

Skills

Technical Tools

- Python, React, MATLAB, LaTeX, C++, HTML, NUMBA, CUDA, WSL
- Numpy, Scikit-learn, PyTorch, TensorFlow, CERN ROOT
- SOLIDWORKS (Mechanical, Electrical, Fluid Dynamics)

Soft Skills

- Communication and team collaboration
- Problem Solving and Experimental Skills

Volunteering

'23-24 Blake's Swim School Swim Coach, Former National Level Swimmer at BREA Aquatics

'25 SLAC National Science Bowl Volunteer

'24 EDUKraine English tutor for Ukrainian refugees

'23 BREA Aquatics events organiser

'23 Fullerton communal parks renovation

Education

University of California, Berkeley **Aug 2024-**
◇ Major in Physics/Applied Mathematics. ◇ Minor in Public Policy
◇ Quantum Computing at Berkeley, hardware subgroup.
◇ Minor in Public Policy ◇ MPS Scholar, Quiz Bowl, Swim Club

Experiences and Projects

Research Assistant at *ARGOS Research Group* **July 2025-**
◇ Part of professor Leo Celi's group at MIT, evaluating biases in AI uses for medicine. Evaluated confirmation biases across multiple databases.

Research Fellow at *Contrary Research* **Jun 2025-**
◇ Contrary is a VC firm driven by talent and research with investments in Andruil, Ramp, Zepto, and Replit.
◇ Produced long form, resource and research backed memos on the latest startups and trends in tech and venture capital.
◇ Currently working on a memo on Ayar Labs, a leader in optical computing, with a brief introduction on the physics of photonics.

Computational Fluid Dynamics at *USC Viterbi* **Jun 2025**
◇ Structured and Unstructured Grid Generation, MPIs and Parallel Computing, CUDA, KOKKOS, and Physics-Informed Deep Learning for fluid dynamics and heat transfer applications.

Nuclear Physics Research at *Berkeley Physics* **Sept-Dec 2024**
◇ Part of the Nuclear Physics Direct Reading Program
◇ Trained machine learning models to identify Higgs Boson signals using Monte Carlo CERN detector data. Achieved ROC Curve of 0.81.
◇ Explored advanced machine learning algorithms and Python libraries for signal analysis. Ran Linux Virtual Machine on WSL for CERN' ROOT Library.

Research/Engineering Intern at *Micro and Nano Manufacturing Lab (UCLA Samueli School of Engineering).* **Jun-Aug 2023**
◇ Research and engineering on drag reduction by MEMS Superhydrophobic (SHPo) surfaces under the guidance of PhD candidate Ray Li.
◇ Designed and 3D-printed testing apparatus for SHPo surfaces to measure bio-fouling capabilities on marine microorganisms.
◇ Gained foundational knowledge of fluid dynamics and micro-electro-mechanical systems.

Founding Engineer at *Petition Power AI* **Dec 2023- Jul 2024**
◇ Petition app that competed in Microsoft's Imagine Cup startup competition, semi-finalist and recipient of $3500 in Azure and OpenAI credits.
◇ Developed Business Plan and Pitch Video, assisted in programming.

Student Lobbyist at *CASC* **Dec 2023- Feb 2024**
◇ Co-wrote two proposals for the state board of education and the Senate Education Committee for the California Association of Student Councils.

Awards, Conferences and Workshops

'25 **Delta,** The Residency
'25 **Nuclear Engineering School,** MTV Consortium
'25 **Tech EE Summit,** Capital One.
'25 **AI Startup School,** Y Combinator.
'25 **Summer Program On Cosmology,** University of Michigan.
'25 **Kissinger Summer Scholar,** Johns Hopkins University.
'24 **Presidential Scholar,** University of Southern California.
'23 **Youth Leader,** Asian Pacific Youth Leadership Project.
'23 **Medalist,** SoCal Swimming Junior Olympics Championships.
'23 **1st/180000** SIFMA Stock Market Simulation Contest.
'22 **Top 33 in GPA** Troy Tech Magnet High School.